Ten Discoveries That Weren't

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Ten celebrated physics claims, from N-rays in 1903 to LK-99 in 2023, collapsed through experimental error rather than fraud or bad luck. Each case below covers the theory that made the claim plausible, the apparatus, and exactly what went wrong.

Introduction: error, not fraud

Physics corrects itself, but the correction is rarely cheap, and it is almost never automatic. In every case in this essay, a competent group with a working instrument announced something new, and the community later agreed that the signal came from the measurement itself rather than from nature. None of these episodes involved fabrication. That is what makes them worth studying: the same failure modes are available to anyone who takes data.

"Experimental error" is used here in a broad sense that covers four families of failure:

1. **Instrument faults**: hardware that did not do what the analysis assumed (OPERA's optical fibre, the TV camera behind a famous false pulsar).
2. **Unmodeled systematics and backgrounds**: a real physical effect that was not the one claimed (Miller's thermal drifts, BICEP2's dust, LK-99's copper sulfide).
3. **Analysis errors**: fitting, selection or significance estimates that manufactured or inflated a signal (Weber's thresholds, the 17 keV kinks, the pentaquark peaks).
4. **Observer bias**: a human detector that reported what it expected (N-rays).

Three kinds of episode are deliberately left out. Fraud, such as Jan Hendrik Schön's organic transistors, Victor Ninov's element 118 or Ranga Dias's hydride superconductors, teaches more about institutions than about measurement. Honest statistical fluctuations reported with appropriate caution, such as the 750 GeV diphoton excess at the LHC in 2015–16, involve no error at all. And live anomalies whose resolution is still argued, such as DAMA/LIBRA's annual modulation, cannot yet be taught as settled history.

The ten cases were chosen because each was presented as a discovery or as strong evidence for new physics, drew serious independent follow-up, and failed for a reason that is now understood. They run in chronological order and span optics, plasma physics, gravitation, nuclear and particle physics, cosmology and condensed matter.

In a 1953 colloquium Irving Langmuir listed the symptoms of what he called pathological science. Paraphrased, they are: an effect at the edge of detectability; a magnitude that does not track the strength of its supposed cause; claims of great accuracy; fantastic theoretical explanations; criticism

met by ad hoc excuses; and a ratio of supporters to critics that rises and then falls away. Several cases below fit that profile closely. Others, notably OPERA, were handled with exemplary caution and were still wrong. The uncomfortable lesson is that good practice lowers the odds of being fooled but never removes them.

1. N-rays (René Blondlot, 1903)

The claim. In 1903 René Blondlot of the University of Nancy, an accomplished experimentalist and corresponding member of the Académie des Sciences, announced a new kind of radiation. He named it the N-ray, after his city. Within about two years more than a hundred scientists had published some 300 papers on N-rays, most of them in France.

The theoretical backdrop. The decade before had been a parade of new radiations: X-rays in 1895, uranium rays in 1896, the electron in 1897, and the alpha, beta and gamma classification soon after. Nothing in the physics of 1903 made one more radiation implausible. Blondlot was studying whether X-rays were polarized, a question that bore directly on whether they were transverse waves.

The experiment. Blondlot's detector was a small electric spark gap placed in the beam of an X-ray tube. He reported that the spark brightened when the gap had certain orientations, which looked like polarization. The effect survived materials that should have blocked X-rays and could be refracted by a quartz or aluminium prism, which X-rays could not. So he concluded that the tube emitted a second, new radiation.

The list of sources grew quickly: Nernst lamps, Welsbach gas mantles, the Sun, heated metals, tempered steel and, according to the physician Augustin Charpentier, the human nervous system. Blondlot later replaced the spark with a faint phosphorescent screen viewed in a darkened room. He dispersed the rays with an aluminium prism and reported a spectrum of discrete lines with measurable "refractive indices."

How it fell apart. Physicists in Germany, Britain and the United States could not see the effect. In September 1904 the American optical physicist Robert W. Wood visited Nancy. During a demonstration of the N-ray spectrum in the dark, Wood quietly removed the aluminium prism from the apparatus. Blondlot went on reading off the same line positions. In another demonstration, a steel file held near the eyes supposedly made a dimly lit clock face easier to see; Wood substituted a piece of wood, and the effect persisted. His short letter in *Nature* ended the phenomenon outside France, though it lingered there for several more years.

What went wrong. The detector was the human eye operating at the threshold of scotopic vision. Near that threshold perceived brightness fluctuates on its own. Rods are absent from the fovea, so a faint object looks brighter with averted gaze, and dark adaptation and fatigue shift the response over tens of minutes. The spark gap was itself unstable from shot to shot. An observer who knows

when the source is present will sort these fluctuations into "signal" and "no signal" without any intent to deceive. Blondlot's photographic evidence did not escape the problem, because the operator who alternated the exposures knew which condition each one was.

The case also shows several of Langmuir's symptoms. The effect's size did not track its supposed cause, since removing the prism changed nothing. And critics were told that some observers simply lacked the sensitivity to see the rays.

The lesson. Wood's prism removal was a blind null test: take away the cause without telling the observer. Every modern blinding protocol descends from that idea, from hidden-signal injections in gravitational-wave searches to the blinded clock frequency in the Fermilab muon g−2 experiment. The deeper point is that no amount of individual expertise substitutes for a detector, human or electronic, that does not know the hypothesis.

2. Ether drift (Dayton Miller, 1921–1933)

The claim. Dayton Miller of the Case School of Applied Science in Cleveland spent three decades repeating the Michelson–Morley experiment. From 1921 onward, working at Mount Wilson Observatory, he reported a small but persistent ether wind of roughly 10 km/s. In 1925 his result won the AAAS prize. His 1933 summary in *Reviews of Modern Physics* went further and inferred an absolute motion of the solar system of about 208 km/s toward the southern constellation Dorado. On first hearing of Miller's result in 1921, Einstein is said to have remarked, "Raffiniert ist der Herrgott, aber boshaft ist er nicht": the Lord is subtle, but not malicious.

The theory. In a stationary ether, light in an arm of length L parallel to the drift velocity v takes longer for the round trip than light in a perpendicular arm, by $\Delta t \approx (L/c)(v/c)^2$. As the interferometer rotates through angle θ, the fringe position oscillates at twice the rotation frequency:

$$N(\theta) = \frac{L}{\lambda}\,\frac{v^2}{c^2}\,\cos 2(\theta - \theta_0), \qquad \Delta N_{90^\circ} = \frac{2L}{\lambda}\,\frac{v^2}{c^2}$$

The azimuth θ_0 points along the drift, and for a cosmic drift it must rotate with sidereal time as the Earth turns. Special relativity predicts no shift at all.

The experiment. Miller's interferometer used a steel cross about 4.3 m across, floated on mercury, with mirrors folding each arm to an effective length of 32 m. For the Earth's orbital speed of 30 km/s, rotating it by 90° should shift the fringes by about 1.1 fringes. The apparatus turned continuously while an observer walked around with it, reading the fringe position at 16 azimuths per turn. Miller's cumulative record exceeded 200,000 readings from more than 12,000 turns. He chose Mount Wilson because he suspected the Earth partially dragged the ether along near the ground, which might explain why lower-altitude experiments had seen so little.

What he found was a second-harmonic signal of about a tenth of a fringe. Taken at face value, that is an ether wind of about 10 km/s.

What went wrong. The instrument was a superb thermometer. A tenth of a fringe at visible wavelengths is an optical path difference of about 57 nm between two 32 m paths. Air's refractive index changes by about 10^{-6} per kelvin, so a difference of roughly 2 mK in mean air temperature between the arms produces the whole effect. Thermal expansion of the steel is even more sensitive: about 0.1 mK between the arms suffices. Miller observed inside a canvas-walled hut on a mountaintop, with sunlight, wind and an observer's body heat all changing through each session.

The data reduction then did the rest. Miller removed fringe drift during each turn by assuming it was linear and averaged many turns together. Real thermal drift is not linear. Folding it at the half-turn period and averaging pushes its residue into exactly the second harmonic the experiment was looking for.

Two reanalyses made this quantitative. Shankland and collaborators, working from Miller's original sheets in 1955, found the periodic effects correlated with temperature conditions and solar time rather than sidereal time. They also noted that Miller's signal was more than ten times smaller than the resolution of an individual reading, recoverable only by averaging hundreds of readings whose errors were not random. In 2006 Thomas Roberts showed that with a proper model of the drift and honest error bars, the data are consistent with no signal.

Meanwhile other experimenters found nothing. Kennedy, Illingworth, Piccard and Stahel, and Michelson himself on Mount Wilson in 1929 all reported null results. In 1930 Georg Joos, with a temperature-stabilized instrument and photographic recording, set an upper limit of about 1.5 km/s. Modern rotating optical-cavity experiments bound anisotropy in the speed of light at the level of 10^{-18}.

The lesson. A modulation technique moves the signal to a known frequency, but slow systematics leak into that frequency too, especially through the analysis. The right test is whether the signal follows the physical variable (sidereal time) or the nuisance (the Sun heating the hut). Averaging beats down random errors only; it does nothing for errors that repeat every turn.

3. Thermonuclear neutrons in ZETA (Harwell, 1958)

The claim. On 25 January 1958, *Nature* published a set of papers from the UK Atomic Energy Authority's Harwell laboratory on ZETA, the Zero Energy Thermonuclear Assembly. The papers reported neutrons from a hot deuterium plasma. At the accompanying press conference, Harwell's director Sir John Cockcroft said he was about 90 percent certain the neutrons were thermonuclear. British newspapers hailed a man-made Sun. The papers themselves were more careful about the neutrons' origin than the press coverage, but the public message was that controlled fusion had begun.

The theory. A current I flowing along a plasma column produces an azimuthal magnetic field that squeezes the column. In equilibrium, the Bennett relation links the current to the line density N and the temperatures:

$$\frac{\mu_0 I^2}{8\pi} = N\,k_B\,(T_e + T_i)$$

Large currents should therefore give hot, dense plasma, and a torus avoids losses from the ends. At ion temperatures of a few million kelvin, a small fraction of deuterons in the Maxwellian tail tunnel through the Coulomb barrier and fuse, half the time via d + d → ^{3}He + n with a 2.45 MeV neutron. Theorists already knew that the simple pinch is violently unstable to "sausage" (m = 0) and kink (m = 1) modes. ZETA added an axial magnetic field and a conducting shell to suppress them.

The experiment. ZETA was an aluminium torus 3 m in major diameter with a 1 m bore, filled with low-pressure deuterium. A capacitor bank drove current pulses of up to about 200 kA in the 1957 runs through a transformer core. From August 1957 the team detected bursts of neutrons at the time of peak current. The yield rose steeply with current, as a thermonuclear yield rising with temperature should. Doppler widths of impurity lines suggested ion temperatures of a few million kelvin.

What went wrong. There is a second way to make D–D neutrons in a pinch. Instabilities that the stabilization did not fully suppress create localized, rapidly changing magnetic fields and hence strong induced electric fields. These accelerate a small population of deuterons to tens of keV along the current direction, and that beam fuses with the cold background plasma. Both mechanisms produce 2.45 MeV neutrons, so counting neutrons cannot tell them apart.

Kinematics can. For a deuteron of energy E_{d} striking a deuteron at rest, the lab-frame neutron energy at angle θ to the deuteron's direction is, to first order in E_{d},

$$E_n(\theta) \simeq E_n^{*} + \sqrt{\frac{m_n}{m_d}\,E_n^{*}E_d}\;\cos\theta, \qquad E_n^{*} \approx 2.45\ \mathrm{MeV}$$

For 20 keV deuterons, forward and backward neutrons differ by about 0.3 MeV, easily measurable. A thermal plasma with no net drift broadens the neutron line symmetrically and shifts it in no particular direction.

In the spring of 1958 Basil Rose, A. E. Taylor and E. Wood at Harwell measured it. Using a high-pressure hydrogen diffusion cloud chamber to record recoil protons, they compared neutrons emitted parallel and antiparallel to the discharge current. The energies differed, and reversing the current reversed the difference. The neutrons came from a small number of fast deuterons, not from a thermonuclear plasma. The Doppler "temperatures" had also been inflated by bulk turbulent motion. Harwell withdrew the thermonuclear interpretation in May 1958. Similar non-thermal neutrons from linear pinches at Los Alamos had already been traced to instabilities.

The lesson. When two mechanisms produce the same particle, the number of particles is not evidence for either one; find the observable that distinguishes them. Here the discriminant was measurable with the technology of the day and was measured only after the announcement. The timing

mattered: the results were released in the months after Sputnik, under Anglo-American agreements on declassifying fusion research, and with enormous political appetite for good news. ZETA had a second life. Its later "quiescent" discharges showed a spontaneously reversed toroidal field that J. B. Taylor explained in 1974 as a relaxed minimum-energy state, the foundation of the reversed-field pinch.

4. Gravitational waves in Weber bars (Joseph Weber, 1969)

The claim. In June 1969 Joseph Weber of the University of Maryland published "Evidence for discovery of gravitational radiation" in *Physical Review Letters*. Two resonant detectors about 1,000 km apart, at Maryland and at Argonne National Laboratory, were registering coincident excitations far more often than chance allowed. In 1970 he reported that the excess peaked at sidereal times when the bars were best oriented toward the Galactic Center. Weber was a serious and inventive physicist who had essentially founded experimental gravitational-wave physics. His claim triggered a worldwide effort to replicate it.

The theory. In general relativity, a gravitational wave is a transverse, traceless strain h in space. It is sourced by a time-varying mass quadrupole and carries an energy flux

$$F = \frac{c^3}{16\pi G}\,\left\langle \dot{h}_+^2 + \dot{h}_\times^2 \right\rangle$$

A passing wave exerts tidal forces on an elastic body. If the wave has power near the body's longitudinal resonance, it rings the mode. Weber's insight was that a massive, high-Q bar acts as a narrowband antenna for kilohertz bursts, such as those expected from stellar collapse.

The experiment. Each detector was an aluminium cylinder roughly 1.5 m long and 0.66 m in diameter, weighing about 1.4 tonnes, suspended in vacuum with acoustic isolation. Its fundamental mode was near 1,660 Hz. Piezoelectric crystals bonded around its waist converted strain into voltage. The amplified, filtered signal was reduced to a record of when the mode energy crossed a threshold. Coincidences between the two sites were then compared with the accidental rate, estimated by shifting one detector's record by various time delays. A real signal shows up as an excess at zero delay.

The thermal noise sets the scale. At room temperature the fundamental mode, with effective mass about half the bar's mass, has an rms amplitude of

$$x_{\mathrm{rms}} = \sqrt{\frac{k_B T}{M_{\mathrm{eff}}\,\omega_0^2}} \approx 2 \times 10^{-16}\ \mathrm{m}$$

so Weber's events corresponded to strains of order 10^{-16}. The first event detected by LIGO in 2015 had a peak strain of 10^{-21}.

Why theorists balked. Signals of that strength, arriving roughly daily from the Galactic Center, imply an enormous energy flux. Dennis Sciama, George Field, Martin Rees and others showed that the Galaxy would have to convert of order a thousand solar masses per year into gravitational radiation. At that rate it could not have survived for its known age. Invoking strong beaming or a very narrow emission band helped only somewhat.

Why experimenters balked. By the mid-1970s groups at Bell Labs (Tony Tyson), IBM (Richard Garwin and James Levine), Glasgow (Ronald Drever), Moscow (Vladimir Braginsky), Munich, Frascati, Rochester and elsewhere had built bars of comparable or better sensitivity. None saw anything. Meanwhile specific problems emerged in Weber's own analysis:

- **Flexible thresholds.** The event threshold and selection algorithm were adjusted after looking at the data. With enough freedom, the zero-delay bin of a coincidence histogram can be made to show an excess from pure noise.
- **The anisotropy flip.** The Earth is transparent to gravitational waves, and a bar's response pattern is symmetric end to end. A Galactic Center source should therefore produce peaks every 12 sidereal hours, not 24. Colleagues recall that Weber's first anisotropy showed a single daily peak, and that a 12-hour pattern appeared after reanalysis once the objection was raised.
- **A time-standard mismatch.** Weber reported coincidences between his detector and the Rochester and Bell Labs detectors. But his timestamps were in local Eastern time and theirs were in Greenwich Mean Time. The "coincident" events were four hours apart, so the analysis was finding signals in data that were uncorrelated by construction.
- **A programming error.** An acknowledged error in Weber's computer analysis had generated spurious coincidences between his own two detectors.

The lesson. Weber never retracted, and he continued to report signals until his death in 2000. His case is a study in how a small excess becomes larger when the analysis is designed while looking at it. Modern gravitational-wave searches still estimate background from time-slides, as Weber did, but they fix the pipeline before unblinding. They also test themselves with hardware injections, including "blind" injections kept secret from the analysts. Weber's legacy is nevertheless real: resonant-mass detection, the first serious engineering of gravitational-wave instruments, and early advocacy of laser interferometry all trace back to him.

5. The 17 keV neutrino (John Simpson, 1985)

The claim. In 1985 John Simpson of the University of Guelph reported a kink in the beta spectrum of tritium. It sat about 17 keV below the 18.6 keV endpoint and was interpreted as emission of a neutrino mass state of 17 keV with a mixing probability of a few percent. Early magnetic-spectrometer searches found nothing, and the idea faded. Then, around 1989–1991, it came back hard. Simpson and Andrew Hime reported kinks in ^{35}S and ^{63}Ni. Hime and N. Jelley at Oxford reported one in ^{35}S, and a Berkeley group reported one in ^{14}C embedded in a germanium detector. The mixing had settled near 1%. For about two years, a heavy neutrino looked like a real possibility.

The theory. If the electron neutrino is a superposition of mass eigenstates, the beta spectrum is a sum of phase-space terms, one per mass. With a light state and a heavy state of mass m_{h} mixed with angle θ,

$$\frac{dN}{dT} \propto F(Z,T)\, p\, E\, (E_0 - T)\Big[\cos^2\theta\,(E_0 - T) + \sin^2\theta\,\sqrt{(E_0 - T)^2 - m_h^2}\;\Theta(E_0 - T - m_h)\Big]$$

where T is the electron's kinetic energy, p and E its momentum and total energy, E_0 the endpoint, and F the Fermi function. The heavy branch switches on below $T = E_0 - m_{\mathrm{h}}$, putting a kink in the Kurie plot at that energy with a fractional size of order $\sin^2\theta$. For tritium the kink sits at 1.6 keV. For ^{35}S, with its 167 keV endpoint, it sits at 150 keV.

A 17 keV neutrino was allowed by the direct limits of the day, since the tau-neutrino bound was tens of MeV. But it demanded new physics. A stable neutrino that heavy would overclose the universe by a factor of more than a hundred ($\Omega h^2 \approx m/94$ eV), so it had to decay. Its Majorana mass contributions also had to evade neutrinoless double-beta-decay limits. Theorists, Sheldon Glashow among them, built models to accommodate it.

What went wrong. The signal was a 1% change in the slope of a smooth spectrum. Measuring it requires knowing the detector's response function, the probability that an electron of energy T' is recorded at T, to much better than 1% across the fitting window. For the solid-state detectors used by most positive experiments, that response has a Gaussian peak and a long low-energy tail. The tail comes from electrons that backscatter out of the crystal, lose energy in dead layers, or scatter off the source backing, collimators and baffles before arriving. An error in modeling the tail shows up as a smooth, percent-level distortion at lower energies, which is exactly what a kink looks like.

The fitting method compounded the problem. Many analyses used a narrow energy window around the expected kink and allowed a free polynomial "shape factor" to absorb unknown distortions. Depending on the window and the polynomial order, such a fit can hide a real kink or create a false one. Positive and negative experiments often disagreed partly because they fit different windows with different shape freedom.

The decisive experiments came in 1992–93. At Argonne, J. Mortara and collaborators used a superconducting solenoid to carry ^{35}S betas to a high-resolution Si(Li) detector. They then did something that should be standard: they mixed a known amount of ^{14}C into the ^{35}S source. The ^{14}C endpoint at 156 keV adds an artificial kink of known size close to where a 17 keV neutrino would put one. Their analysis found the planted kink and found no 17 keV neutrino, with $\sin^2\theta = -0.0004 \pm 0.0008$ (stat) ± 0.0008 (syst). A Tokyo group reached a similar null result with ^{63}Ni. Andrew Hime then showed that electron scattering in his own Oxford apparatus could reproduce the distortion he had reported, and said so publicly.

The lesson. Agreement among several experiments is not independent confirmation if they share the same class of systematic error. Here, most positive results came from solid-state detectors with similar response tails, while most negative ones came from magnetic spectrometers. The Argonne

mixed-source test is a model for demonstrating sensitivity: inject a known signal of the expected shape and show the analysis recovers it. The episode, reconstructed in detail by the historian Allan Franklin, is also a caution about analyses whose flexibility is comparable to the size of the effect.

6. Cold fusion (Martin Fleischmann and Stanley Pons, 1989)

The claim. On 23 March 1989, at a press conference at the University of Utah, the electrochemists Martin Fleischmann and Stanley Pons announced nuclear fusion in a tabletop electrolysis cell. They reported excess heat far beyond any chemical source, along with neutrons and tritium. Fleischmann was a Fellow of the Royal Society and one of the leading electrochemists of his generation. Within weeks, hundreds of laboratories were trying to reproduce the result. Separately, Steven Jones at Brigham Young University reported a much weaker neutron signal from similar cells, at a level some thirteen orders of magnitude below what the Utah heat would require.

The theory, and why it forbids this. Deuterium fusion proceeds through three channels:

$$
\begin{aligned}
d + d &\to {}^{3}\mathrm{He}\,(0.82\ \mathrm{MeV}) + n\,(2.45\ \mathrm{MeV}) && \approx 50\% \\
d + d &\to t\,(1.01\ \mathrm{MeV}) + p\,(3.02\ \mathrm{MeV}) && \approx 50\% \\
d + d &\to {}^{4}\mathrm{He} + \gamma\,(23.8\ \mathrm{MeV}) && \sim 10^{-7}
\end{aligned}
$$

At room temperature, fusion requires tunneling through a Coulomb barrier of a few hundred keV with a relative energy of about 0.03 eV. Even in the D_2 molecule, where the nuclei sit 0.74 Å apart, Koonin and Nauenberg calculated a fusion rate of order 10^{-64} per molecule per second. Loading deuterium into palladium does not help geometrically. In PdD, deuterons occupy octahedral interstitial sites about 2.9 Å apart, farther than in D_2.

The experiment. Each cell was a glass Dewar containing 0.1 M LiOD in heavy water, with a palladium cathode (rod, sheet or cube) and a platinum anode. Constant-current electrolysis ran for days to weeks to load the palladium with deuterium. A thermistor measured the electrolyte temperature, and a resistive heater provided calibration. Excess power was computed as the heat inferred from the temperature minus the electrical input, less the enthalpy carried away as D_2 and O_2 gas: the input term was (V − 1.54 V) × I, where 1.54 V is the thermoneutral potential for D_2O. Some electrodes reportedly produced several watts per cubic centimetre of palladium.

What went wrong. Three independent lines of evidence collapsed.

1. **The byproducts were missing.** One watt of d–d fusion requires about 1.7 × 10^{12} reactions per second and would emit nearly 10^{12} neutrons per second. That flux would have delivered a lethal dose to anyone standing near the cell in well under an hour. The neutron rates actually claimed were smaller by a factor of order 10^{8}, and tritium was similarly deficient. Proponents' later proposal, that the heat came from ^{4}He production without the 23.8 MeV gamma ray, requires a mechanism for coupling MeV-scale energy directly into lattice phonons. No such mechanism is known.

2. **The calorimetry was fragile.** The cells were open, so the energy bookkeeping assumed all D_2 and O_2 escaped. Any recombination of the gases on the electrode surfaces inside the cell releases up to 1.54 V × I as heat, which appears as "excess"; at 0.5 A that is 0.77 W. The cells were not mechanically stirred, so a single thermistor sampled a temperature field with gradients. The calibration's cell constant also drifted as electrolysis consumed the heavy water and the liquid level fell. Nathan Lewis's group at Caltech repeated the measurement with stirred cells and careful calibration and found no excess heat. They argued that the Utah numbers were consistent with calibration artifacts.
3. **The gamma line was not a gamma line.** Fleischmann and Pons showed a spectrum with a peak attributed to the 2.224 MeV gamma ray from neutron capture on protons in the surrounding water. Richard Petrasso's MIT group noted that the peak was narrower than the sodium-iodide detector's resolution permits at that energy. It also lacked the Compton edge near 2.0 MeV that any real line of that energy must produce, and its energy changed between preliminary and published versions of the data. The reported tritium had a mundane explanation too: electrolysis preferentially evolves the lighter isotopes, concentrating trace tritium already present in the heavy water.

A U.S. Department of Energy panel concluded in November 1989 that there was no convincing evidence for fusion. A second DOE review in 2004 reached essentially the same conclusion.

The lesson. When a claimed process has unavoidable byproducts, check the ratio of signal to byproduct before believing the signal. The neutron-to-heat ratio here was off by eight orders of magnitude, and no calorimetric subtlety can overcome that. Calorimetry is harder than it looks, and "excess heat" is a difference of two large numbers, each with its own systematics. Finally, announcing a result at a press conference, before peer review and before sharing technical details, left the rest of the community replicating against a moving target.

7. The Θ^+ pentaquark (LEPS and others, 2003)

The claim. In 2003 the LEPS collaboration at the SPring-8 synchrotron in Japan reported a narrow peak at 1.54 GeV in the K^+n invariant mass. The quantum numbers had to be baryon number +1 and strangeness +1, which no three-quark state can carry. The minimal content is uudd $\bar{s}$: a pentaquark. Within a year, about ten experiments using entirely different beams and targets reported the same state. They included DIANA at ITEP, CLAS at Jefferson Lab, SAPHIR at Bonn, HERMES and ZEUS at DESY, COSY-TOF, and SVD at Protvino. Individual significances ranged from about 4 to 7σ. The Particle Data Group listed the $\Theta^+(1540)$ with three stars in 2004.

The theory. Exotic baryons had been sought since the 1960s without success. What made 2003 different was a sharp prediction. In 1997 Diakonov, Petrov and Polyakov, using the chiral quark soliton model, predicted an antidecuplet of baryons whose lightest member had strangeness +1, a mass near 1530 MeV and an unusually narrow width below 15 MeV. LEPS explicitly tested this prediction and found a state close to it. After the announcement, Jaffe and Wilczek offered a diquark picture, and hundreds of theory papers followed.

The experiment. LEPS produced 1.5–2.4 GeV photons by backscattering laser light off the 8 GeV electron beam. It studied $\gamma n \to K^- K^+ n$ on neutrons in a carbon-containing target, detecting the kaon pair in a forward spectrometer and reconstructing the K^+n mass with a correction for Fermi motion. CLAS studied $\gamma d \to K^+ K^- pn$ and reported 5.2σ. Other experiments looked at $K^0 p$ or $K^+ n$ combinations in very different kinematics.

What went wrong. High-statistics searches found nothing. BaBar, Belle, BES, HERA-B, CDF, FOCUS, HyperCP, SPHINX and the LEP experiments saw no Θ^+, often with vastly larger samples. The decisive blow came from CLAS itself. In a dedicated 2004 run on deuterium with several times the original statistics, published in 2006, the 5.2σ peak was gone. A matching proton run found no trace of the SAPHIR signal in the same reaction. No single hardware fault explains the dozen false positives. Instead, the same analysis weaknesses recurred:

- **Background shapes.** In low-statistics spectra, the background under a narrow peak was typically fit with a smooth function chosen by the analysts. A background drawn slightly too low under the target mass inflates the signal.
- **Kinematic reflections.** In final states such as K^+K^-n, meson resonances decaying to K^+K^- (for example the a_2 and f_2 tensor mesons) and cuts to remove the φ can sculpt structures in the K^+n mass distribution.
- **Cuts tuned while looking.** Selection criteria were often optimized with knowledge of where the peak should be.
- **Optimistic significance.** Many papers quoted $S/\sqrt{B}$, ignoring the uncertainty on B and the number of places a peak could have appeared.

The last point is worth making quantitative. With a background estimate B carrying its own uncertainty σ_B,

$$Z_{\text{naive}} = \frac{S}{\sqrt{B}} \qquad \text{vs.} \qquad Z \approx \frac{S}{\sqrt{B + \sigma_B^2}}$$

For S = 40 and B = 40 events, the naive significance is 6.3σ. If the background is known only to ±10 events, the significance drops to 3.4σ. Suppose the peak could have appeared anywhere in a window about 20 resolution widths wide. The trials factor then raises the p-value about twentyfold, leaving a global significance near 2.5σ.

There were physical red flags too. The reported masses scattered from about 1520 to 1555 MeV, more than the quoted errors allowed. Existing K^+n and K^+d scattering data required the width to be of order 1 MeV or less, which implies weak coupling to KN. A state that weakly coupled should have been produced far less copiously than the peaks suggested. By 2008 the PDG had removed the Θ^+ from its tables, calling the whole story a curious episode in the history of science.

The lesson. Ten observations at 4–7σ each sound overwhelming. But if each carries an unaccounted-for systematic of similar size and direction, and each was analyzed by people looking for a peak at a predicted mass, the ten are not independent. The pentaquark episode was a major reason blind analysis became routine in nuclear and particle physics. Genuine pentaquarks were eventually

found: in 2015 and 2019 LHCb observed hidden-charm P_c states in $\Lambda_b \to J/\psi\ K^-\ p$ decays. That analysis used tens of thousands of decays and a full amplitude analysis, not a bump on a hand-drawn background.

8. Superluminal neutrinos (OPERA, 2011)

The claim. On 23 September 2011, the OPERA collaboration presented a measurement at CERN showing that muon neutrinos travelling 730 km through the Earth from CERN to the Gran Sasso laboratory arrived early:

$$\delta t = 60.7 \pm 6.9\,(\text{stat}) \pm 7.4\,(\text{sys})\ \text{ns}, \qquad \frac{v - c}{c} = (2.48 \pm 0.28 \pm 0.30) \times 10^{-5}$$

That is a 6.0σ departure from the speed of light. OPERA's conduct was careful. The collaboration explicitly declined to claim a discovery, posted everything on arXiv, and asked the community to find the mistake. The result was still widely reported as a possible violation of relativity.

The experiment. OPERA was built to see $\nu_\mu \to \nu_\tau$ oscillation in the CNGS beam, using 1.25 kilotonnes of lead-emulsion bricks. Timing was a by-product. 400 GeV protons from the SPS struck a graphite target in two 10.5 μs extractions per cycle. The resulting mesons decayed to neutrinos with a mean energy of about 17 GeV. A beam current transformer recorded each proton waveform, and the sum of those waveforms gave the expected time profile of neutrino arrivals. About 16,000 neutrino events recorded at Gran Sasso from 2009 to 2011 were fit to that profile by maximum likelihood, with the time offset δt free. Clocks at the two sites were synchronized by GPS common-view time transfer with caesium standards, calibrated by the Swiss and German metrology institutes. The baseline was surveyed geodetically to 20 cm.

In late 2011, CERN ran a special beam with 3 ns proton bunches spaced 524 ns apart. Each neutrino event could now be matched to its bunch directly, with no statistical waveform fit. Twenty events gave δt = 62.1 ± 3.7 ns, apparently confirming the effect.

Why theorists doubted it. Three independent arguments made the result nearly impossible to accept:

- **SN 1987A.** Kamiokande-II, IMB and Baksan detected about two dozen 10–40 MeV antineutrinos from a supernova 168,000 light-years away. They arrived about three hours before the light, as expected from the time the shock takes to break out of the star. A speed excess of 2.5×10^{-5} would have brought them about four years early. The observed timing bounds $|v - c|/c$ at about 2×10^{-9}, though at energies a thousand times lower than OPERA's.
- **Tachyonic mass.** A tachyon with $E^2 = p^2c^2 - \mu^2c^4$ has $(v - c)/c \approx \mu^2c^4/2E^2$. At 17 GeV the OPERA number requires $|\mu| \approx 120$ MeV, which is absurd given the sub-eV neutrino mass limits. OPERA also saw no energy dependence in δt.
- **Cohen–Glashow pair emission.** A neutrino moving faster than light in a Lorentz-violating vacuum can radiate e^+e^- pairs, $\nu \to \nu\ e^+\ e^-$, above a threshold of about $2m_e/\sqrt{\delta} \approx 140$ MeV. Andrew

Cohen and Sheldon Glashow computed the energy-loss rate as

$$\frac{dE}{dx} = -\frac{25}{448}\,\frac{G_F^2}{192\pi^3}\,E^6\,\delta^3, \qquad \delta \equiv \frac{v_\nu^2}{c^2} - 1$$

Over 730 km this would have degraded every neutrino above about 12.5 GeV, reshaping the beam spectrum and producing showers of pairs. The ICARUS detector, in the same hall, looked for exactly this and found neither.

What went wrong. In February 2012 OPERA announced two problems in its timing chain. The GPS receiver sat outside the mountain, and its timing signal travelled 8.3 km by optical fibre to the underground master clock. There, the fibre's connector to the clock's electronics card was not fully seated. The reduced light level delayed the card's detection of each synchronization pulse, so OPERA's clock ran late and every neutrino appeared to arrive early. A second, smaller problem, in the oscillator that interpolated timestamps between GPS synchronizations, pushed in the opposite direction.

The cleanest confirmation came from an independent reference. The neighbouring LVD detector recorded nearly horizontal cosmic-ray muons that crossed both detectors. Comparing timestamps for these muons over the years showed an offset of 73 ± 9 ns in OPERA's clock. It began in mid-2008 and disappeared when the connector was reseated in December 2011. ICARUS measured δt = 0.3 ± 4.9 (stat) ± 9.0 (sys) ns with the bunched beam. OPERA's corrected final result was δt = 6.5 ± 7.4 (stat) (+8.3, −8.0) (sys) ns, consistent with zero. The spokesperson, Antonio Ereditato, resigned in March 2012.

The lesson. Repetition is not replication. The bunched-beam run removed every statistical concern while reusing the same faulty clock, so it reproduced the error to within 2 ns. A systematic that is stable across a whole dataset is invisible to internal checks; only an independent reference, like LVD's muons, can reveal it. The mundane end of the apparatus, connectors and cables and firmware, deserves the same end-to-end calibration as the exotic end, and it should be repeated over the life of the experiment. OPERA also shows that doing everything right in communication does not protect against being wrong. It does make the correction faster and less painful.

9. Primordial B-modes (BICEP2, 2014)

The claim. On 17 March 2014, at a press conference at the Harvard-Smithsonian Center for Astrophysics, the BICEP2 collaboration announced B-mode polarization in the cosmic microwave background at degree angular scales. They attributed it to gravitational waves from inflation, with a tensor-to-scalar ratio r = 0.20 (+0.07, −0.05) and r = 0 excluded at 7.0σ. If true, it was the first direct evidence for inflation, a measurement of its energy scale, and indirect evidence for quantized gravity. A video of the experimenters telling Andrei Linde at his front door circulated widely.

The theory. CMB polarization is generated by Thomson scattering of a quadrupolar radiation field at last scattering. A polarization map can be decomposed into a curl-free E-mode pattern and a divergence-free B-mode pattern, which have opposite parity. Scalar density perturbations produce

only E-modes at linear order. Tensor perturbations, that is, gravitational waves, produce both. So a primordial B-mode signal peaking near multipole $\ell \approx 80$ is a clean signature of inflationary gravitational waves. Gravitational lensing also converts E into B, but mainly at smaller scales ($\ell \sim 1000$). In single-field slow-roll inflation, r fixes the energy scale of the inflaton potential:

$$V^{1/4} \approx 1.04 \times 10^{16}\ \mathrm{GeV} \left(\frac{r}{0.01}\right)^{1/4}$$

An r of 0.2 also implies, through the Lyth bound, that the inflaton moved over a super-Planckian field range. That would have favored simple large-field models such as $m^2\varphi^2$ inflation, which predicts r ≈ 0.13–0.16.

The experiment. BICEP2 was a small-aperture (26 cm) refracting telescope at the South Pole with 512 antenna-coupled transition-edge-sensor bolometers operating at a single frequency, 150 GHz. From 2010 to 2012 it mapped about 380 square degrees of the "Southern Hole," a field chosen because it has among the lowest Galactic dust emission in the sky. The B-mode power detected at $50 \lesssim \ell \lesssim 150$ was well above the lensing expectation.

What went wrong. The sky signal was real; the Keck Array later confirmed it at 150 GHz. The error was in attributing it to the early universe rather than to the Milky Way.

Interstellar dust grains are elongated and align with the Galactic magnetic field. Their thermal emission is therefore polarized, with a polarization fraction that can reach about 20%. Dust emission follows a modified blackbody (β ≈ 1.6, T ≈ 20 K) and rises steeply with frequency, while the CMB has a blackbody spectrum. Separating the two requires at least two frequencies. At a single frequency, BICEP2 had to rely on external dust models to estimate the contamination in its field. Several of those models were extrapolations from pre-Planck data with an assumed polarization fraction. The most important one was built partly from a preliminary Planck dust-polarization map that the team digitized from a conference presentation slide. It was unclear whether that map was appropriate for this purpose; critics argued that it could understate the polarization fraction.

The team's one spectral handle was a cross-correlation with 100 GHz maps from its predecessor, BICEP1. That comparison disfavored a pure-dust origin at only 1.7σ. Within weeks, Flauger, Hill and Spergel, and separately Mortonson and Seljak, showed that plausible dust levels could explain the entire signal. The published paper, in June 2014, added that the dust models were not constrained well enough to exclude that possibility.

Planck's 353 GHz polarization maps, released in September 2014, settled it. Even in the cleanest regions, dust B-mode power extrapolated to 150 GHz was comparable to the whole BICEP2 signal. A joint BICEP2/Keck/Planck analysis in early 2015 detected dust at high significance and found r < 0.12 at 95% confidence, consistent with zero. Today's BICEP/Keck limit is r < 0.036 (95%), from multifrequency data through 2018. That rules out $m^2\varphi^2$ inflation and pushes the search toward r ~ 10^{-3} with next-generation experiments.

The lesson. A foreground is a systematic in its own right, and a single-frequency experiment cannot measure it. Confidence in a subtraction is only as good as the least-validated input, here a digitized slide of preliminary data. The significance BICEP2 quoted was statistical. The dominant uncertainty, the dust amplitude in its field, was a modeling assumption that the statistics did not include. The press conference before peer review turned a reasonable preprint into a public reversal.

10. LK-99 (2023)

The claim. On 22 July 2023, two preprints from Sukbae Lee, Ji-Hoon Kim, Young-Wan Kwon and collaborators at the Quantum Energy Research Centre in Seoul claimed the first room-temperature, ambient-pressure superconductor. The material, called LK-99, was a copper-doped lead apatite, $Pb_{10-x}Cu_x(PO_4)_6O$ with $x \approx 1$. The claimed critical temperature was 400 K (127 °C) or higher. The evidence was a sharp drop in resistivity on cooling near 380 K, current–voltage curves resembling a critical current, a heat-capacity anomaly, and a video of a small flake partially levitating over a magnet. The synthesis looked simple enough for a well-equipped undergraduate lab, and within days dozens of groups, and many amateurs, were trying it.

The theory. The authors proposed that copper substituting for lead shrinks the lattice slightly, and that the resulting strain creates superconducting "quantum wells" along the apatite's columns. A density-functional calculation posted within a week found isolated, very flat bands near the Fermi level derived from copper d-orbitals. Flat bands mean a high density of states and strong correlations, both ingredients that theorists associate with unconventional superconductivity, and the calculation was widely read as support. The more sober reading was that a narrow, partially filled band of localized d-states, once correlations are included, usually signals a Mott or charge-transfer insulator rather than a superconductor. Later calculations with realistic structures and correlation effects found just that.

The experiment. LK-99 was made in two steps. First lanarkite, $Pb_2(SO_4)O$, and copper phosphide, Cu_3P, were prepared. They were then ground together and reacted at 925 °C in an evacuated quartz tube. Resistivity was measured on the resulting polycrystalline pellets, magnetic response on pieces of them.

What went wrong. The sulfur in the lanarkite had to go somewhere. Much of it ended up as copper(I) sulfide, Cu_2S, a minority phase that is hard to spot in powder X-ray diffraction but threads through the pellets. Cu_2S has a first-order structural transition between a high-temperature hexagonal phase and a low-temperature monoclinic phase near 380 K. Across it, the resistivity changes by three to four orders of magnitude.

Prashant Jain at Illinois pointed out that the temperature matched LK-99's "transition" almost exactly. A group at the Institute of Physics of the Chinese Academy of Sciences then measured it. They found Cu_2S resistivity falling from 13.7 Ω·cm at 385.6 K to 0.006 Ω·cm at 381.5 K, with the thermal hysteresis of a first-order transition. LK-99 samples containing Cu_2S reproduced the shape and tem-

perature of the published resistivity drop. A superconducting transition in zero field shows no such hysteresis, and the "superconducting" resistivity in the original data never convincingly reached zero.

The magnetic evidence failed too. Pure single crystals of $Pb_{10-x}Cu_x(PO_4)_6O$, grown by the floating-zone method at the Max Planck Institute in Stuttgart, were transparent purple insulators. Several groups reproduced the "half levitation" with impure flakes and traced it to ferromagnetic and diamagnetic impurity phases. A flake with one edge resting on the magnet is what magnetic torque on ferromagnetic inclusions produces. A Meissner state expels flux from the whole sample and, with pinning, can hold it fully suspended. Less than a month after the preprints, a *Nature* news headline could state flatly that LK-99 was not a superconductor.

The lesson. Superconductivity has a demanding definition, and each part is a separate test:

- Zero DC resistance, measured with four probes and current reversal, not merely a drop.
- Bulk flux expulsion, with zero-field-cooled and field-cooled magnetization showing a shielding fraction consistent with the sample volume.
- Thermodynamic consistency: a specific-heat anomaly at T_c and suppression of T_c by a magnetic field.

In a multiphase polycrystal, a resistance drop can come from percolation through a minority phase with its own transition, so phase purity is part of the evidence, not a detail. The field has a name for these episodes, USOs (unidentified superconducting objects), because they recur. LK-99 was an honest mistake of this kind, unlike the retracted hydride claims of the same year that are excluded here as misconduct. It was also the fastest correction in this essay: open preprints and dozens of rapid replications reached a consensus in about three weeks.

Summary

Every false signal had a mundane cause. In most cases a specific, feasible test would have exposed it before the announcement.

#	Case	Claim	Stated confidence	Actual cause	What exposed it
1	N-rays (1903)	New radiation from lamps, metals, the Sun, nerves	Reproduced by 100+ scientists	Observer bias at the threshold of vision; no blinding	Wood removed the prism unseen (1904)
2	Miller ether drift (1921–33)	Ether wind of ~10 km/s	~0.1 fringe from 200,000+ readings	mK thermal gradients; drift folded into the second harmonic	Null results (Joos, < 1.5 km/s); Shankland reanalysis (1955)
3	ZETA (1958)	Thermonuclear D–D fusion	"90% certain" (Cockcroft, to the press)	Beam–target neutrons from instability-driven deuterons	Neutron energy asymmetry that flipped with current (Rose et al., 1958)
4	Weber bars (1969)	Gravitational-wave bursts from the Galactic Center	Excess zero-delay coincidences	Post hoc thresholds; time-standard and code errors	Null results from ~8 labs; Galactic energy budget

#	Case	Claim	Stated confidence	Actual cause	What exposed it
5	17 keV neutrino (1985)	Heavy neutrino with ~1% mixing	Several positive experiments	Detector response tails and scattering; narrow-window fits	Argonne mixed $^{35}S/^{14}C$ source test (1993); Hime's scattering analysis
6	Cold fusion (1989)	Room-temperature d–d fusion	Watts of excess heat, plus neutrons and tritium	Calorimetry artifacts; misidentified gamma line; tritium enrichment	Neutron-to-heat ratio $\sim 10^{8}$ too small; stirred-cell calorimetry
7	Θ^{+} pentaquark (2003)	Exotic uudd $\bar{s}$ baryon at 1540 MeV	4–7σ in ~10 experiments	Background shapes, reflections, tuned cuts, trials factor	High-statistics repeats (CLAS 2006, BaBar, Belle)
8	OPERA (2011)	Neutrinos faster than light	6.0σ	Badly seated timing fibre (73 ns)	LVD–OPERA cosmic-muon timing; ICARUS
9	BICEP2 (2014)	Inflationary B-modes, r = 0.20	r = 0 excluded at 7.0σ	Polarized Galactic dust underestimated	Planck 353 GHz maps; joint BICEP2/Keck/Planck analysis (2015)
10	LK-99 (2023)	Superconductor with $T_c \geq 400$ K	Resistance drop, partial levitation	Cu_2S transition near 380 K; magnetic impurities	Cu_2S measurements; insulating single crystals

Lessons

The ten cases span 120 years and most of physics, yet they fail in a small number of recurring ways. Each failure mode now has a standard countermeasure, usually adopted because of one of these episodes.

1. **Blind the observer and the analysis.** Blondlot's eye, Weber's thresholds, the pentaquark cuts and Miller's reductions were all shaped by knowing what the answer should be. Modern practice hides the signal region, an offset or a calibration constant until the analysis is frozen. Examples include LIGO's blind hardware injections, the hidden clock frequency in muon g−2, and the fake "salt" events planted in dark-matter searches.
2. **Prove sensitivity with a known signal.** Mortara's ^{14}C admixture and Wood's missing prism are the same idea. Inject something of the expected shape, or remove the supposed cause, and see whether the analysis responds correctly.
3. **Repetition is not replication.** OPERA's bunched beam reproduced a faulty clock. Solid-state beta spectrometers reproduced one another's response tails. BICEP2's own later data reproduced the dust. Confirmation counts only when it comes with independent systematics: a different detector technology, a different frequency, or an independent time reference.
4. **Check the unavoidable companions.** Fusion heat without neutrons, galactic gravitational radiation without galactic mass loss, and superluminal neutrinos without pair emission or a supernova precursor all failed a budget that nature enforces. These checks are not deference to theory. They are other measurements that the new claim must also satisfy.
5. **Find the discriminating observable before announcing.** ZETA's neutron anisotropy, Miller's sidereal-versus-solar test, BICEP2's frequency spectrum and LK-99's Meissner fraction all dis-

tinguish the exciting explanation from the mundane one. In each case the test was feasible, and in several it was run only after the announcement.

6. **A significance is conditional on its background model.** OPERA quoted 6.0σ, BICEP2 7.0σ, and CLAS 5.2σ. All three were statistically correct and physically wrong. A p-value says nothing about a systematic it was not told about. Uncertainty in the background and the trials factor belong in the quoted number.
7. **Know where your systematics live.** They cluster at the modulation harmonic, at the threshold, at the edges of the fitting window, in minority phases, and in the connectors nobody thinks about.
8. **Communicate in proportion to what you know.** ZETA, cold fusion and BICEP2 were announced at press conferences before independent scrutiny. OPERA and LK-99 were posted openly and corrected within weeks to months. The physics was equally wrong, but the second route cost the field far less.

It is also worth noticing who comes out of these stories well. Andrew Hime publicly explained his own false signal. OPERA found its own loose connector. In a case that nearly made this list, Andrew Lyne stood up at a 1992 American Astronomical Society meeting to withdraw his group's claim of a planet around pulsar PSR 1829−10. The spurious six-month signal came from an error in correcting for the Earth's orbit. The audience gave him a standing ovation.

Other cases that nearly made the list. Price and collaborators' 1975 moving magnetic monopole, recorded in a balloon-borne detector stack, was more plausibly a heavy cosmic-ray nucleus that fragmented. A 1989 half-millisecond optical pulsar in SN 1987A turned out to be electronic interference from a television camera at the telescope. Positive "fifth force" results from towers and mines in the late 1980s were traced to unmodeled local mass distributions, while torsion balances found nothing. The ITEP claim of a roughly 30 eV electron-neutrino mass in 1980 failed on molecular final-state and resolution effects; KATRIN now limits the mass to below 0.45 eV. And UA1's 1984 hint of a top quark near 40 GeV dissolved with better background estimates, 11 years before the top was found near 175 GeV.

Richard Feynman's advice to Caltech graduates in 1974 is the shortest summary of all ten cases: the first principle is not to fool yourself, and you are the easiest person to fool. Every group here had good instruments and good people. What each lacked, at the critical moment, was a test designed by someone trying to prove them wrong.

Further reading

These are the primary papers and retrospectives behind each case. Linked items were checked while this essay was written; the others are standard citations given from memory, so verify page numbers before citing them.

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